$\{+b_1, b_2, b_3\}/a$	Gauge-link paths	$\{-b_1, b_2, b_3\}/a$	Gauge-link paths
{1, 1, 0}	{{2, 1}, {1, 2}}	{-1, 1, 0}	{{2, -1}, {-1, 2}}
{2, 2, 0}	{{2, 1, 2, 1}, {1, 2, 1, 2}}	{-2, 2, 0}	{{2, -1, 2, -1}, {-1, 2, -1, 2}}
{3, 3, 0}	{{2, 1, 2, 1, 2, 1}, {1, 2, 1, 2, 1, 2}}	{-3, 3, 0}	{{2, -1, 2, -1, 2, -1}, {-1, 2, -1, 2, -1, 2}}
{4, 4, 0}	{{2, 1, 2, 1, 2, 1, 2, 1}, {1, 2, 1, 2, 1, 2, 1, 2}}	{-4, 4, 0}	{{2, -1, 2, -1, 2, -1, 2, -1}, {-1, 2, -1, 2, -1, 2, -1, 2}}
{5, 5, 0}	{{2, 1, 2, 1, 2, 1, 2, 1, 2, 1}, {1, 2, 1, 2, 1, 2, 1, 2, 1, 2}}	{-5, 5, 0}	{{2, -1, 2, -1, 2, -1, 2, -1, 2, -1}, {-1, 2, -1, 2, -1, 2, -1, 2, -1, 2}}
{1, 2, 0}	{{2, 1, 2}}	{-1, 2, 0}	{{2, -1, 2}}
{2, 4, 0}	{{2, 1, 2, 2, 1, 2}}	{-2, 4, 0}	{{2, -1, 2, 2, -1, 2}}
{3, 6, 0}	{{2, 1, 2, 2, 1, 2, 2, 1, 2}}	{-3, 6, 0}	{{2, -1, 2, 2, -1, 2, 2, -1, 2}}
{2, 1, 0}	{{1, 2, 1}}	{-2, 1, 0}	{{-1, 2, -1}}
{4, 2, 0}	{{1, 2, 1, 1, 2, 1}}	{-4, 2, 0}	{{-1, 2, -1, -1, 2, -1}}
{6, 3, 0}	{{1, 2, 1, 1, 2, 1, 1, 2, 1}}	{-6, 3, 0}	{{-1, 2, -1, -1, 2, -1, -1, 2, -1}}
{4, 1, 0}	{{1, 1, 2, 1, 1}}	{-4, 1, 0}	{{-1, -1, 2, -1, -1}}
{8, 2, 0}	{{1, 1, 2, 1, 1, 1, 1, 2, 1, 1}}	{-8, 2, 0}	{{-1, -1, 2, -1, -1, -1, -1, 2, -1, -1}}
{1, -1, 0}	{{-2, 1}, {1, -2}}	{-1, -1, 0}	{{-2, -1}, {-1, -2}}
{2, -2, 0}	{{-2, 1, -2, 1}, {1, -2, 1, -2}}	{-2, -2, 0}	{{-2, -1, -2, -1}, {-1, -2, -1, -2}}
{3, -3, 0}	{{-2, 1, -2, 1, -2, 1}, {1, -2, 1, -2, 1, -2}}	{-3, -3, 0}	{{-2, -1, -2, -1, -2, -1}, {-1, -2, -1, -2, -1, -2}}
{4, -4, 0}	{{-2, 1, -2, 1, -2, 1, -2, 1}, {1, -2, 1, -2, 1, -2, 1, -2}}	{-4, -4, 0}	{{-2, -1, -2, -1, -2, -1, -2, -1}, {-1, -2, -1, -2, -1, -2, -1, -2}}
{5, -5, 0}	{{-2, 1, -2, 1, -2, 1, -2, 1, -2, 1}, {1, -2, 1, -2, 1, -2, 1, -2, 1, -2}}	{-5, -5, 0}	{{-2, -1, -2, -1, -2, -1, -2, -1, -2, -1}, {-1, -2, -1, -2, -1, -2, -1, -2, -1, -2}}
{1, -2, 0}	{{-2, 1, -2}}	{-1, -2, 0}	{{-2, -1, -2}}
{2, -4, 0}	{{-2, 1, -2, -2, 1, -2}}	{-2, -4, 0}	{{-2, -1, -2, -2, -1, -2}}
{3, -6, 0}	{{-2, 1, -2, -2, 1, -2, -2, 1, -2}}	{-3, -6, 0}	{{-2, -1, -2, -2, -1, -2, -2, -1, -2}}
{2, -1, 0}	{{1, -2, 1}}	{-2, -1, 0}	{{-1, -2, -1}}
{4, -2, 0}	{{1, -2, 1, 1, -2, 1}}	{-4, -2, 0}	{{-1, -2, -1, -1, -2, -1}}
{6, -3, 0}	{{1, -2, 1, 1, -2, 1, 1, -2, 1}}	{-6, -3, 0}	{{-1, -2, -1, -1, -2, -1, -1, -2, -1}}
{4, -1, 0}	{{1, 1, -2, 1, 1}}	{-4, -1, 0}	{{-1, -1, -2, -1, -1}}
{8, -2, 0}	{{1, 1, -2, 1, 1, 1, 1, -2, 1, 1}}	{-8, -2, 0}	{{-1, -1, -2, -1, -1, -1, -1, -2, -1, -1}}